

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,0} [m_2, \tilde{\mu}] \delta_{i1i2} - \right. \\ & \frac{1}{6} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} - \\ & \frac{1}{6} \tilde{\mu} g_2^2 \overline{y_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} + \frac{3}{8} \tilde{\mu} g_2^2 \overline{y_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} - \\ & \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} + \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] \delta_{i1i2} - \\ & \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_u^{\text{r}}] \delta_{i1i2} - \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} - \\ & \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} + \frac{3}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} - \\ & \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} + \frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] \delta_{i1i2} - \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,1,0} [\tilde{\mu}, m_2] \delta_{i1i2} - \\ & \frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] \delta_{i1i2} + \frac{7}{8} m_2 \tilde{\mu} g_2^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] \delta_{i1i2} - \\ & \frac{1}{2} m_2 \tilde{\mu} g_2^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_d^{\text{p}}, m_q^{\text{i2}}] + \\ & \frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, m_d^{\text{p}}, m_q^{\text{i2}}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_d^{\text{p}}, m_q^{\text{i2}}] + \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_d^{\text{p}}, \tilde{\mu}] - \\ & \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_d^{\text{p}}, \tilde{\mu}] - \frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, m_q^{\text{i2}}, m_d^{\text{p}}] - \\ & \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_q^{\text{i2}}, m_u^{\text{p}}] + \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \\ & \text{LF}_{2,2,1,-1} [m_1, m_q^{\text{i2}}, m_u^{\text{p}}] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_q^{\text{i2}}, m_u^{\text{p}}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, m_u^{\text{p}}, m_q^{\text{i2}}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_u^{\text{p}}, \tilde{\mu}] - \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_u^{\text{p}}, \tilde{\mu}] - \frac{5}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{\text{p}}] + \\ & \frac{1}{48} m_1 \tilde{\mu} g_1^2 (-\overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} + \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}}) \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{\text{i2}}] - \\ & \frac{7}{48} m_1 \tilde{\mu} g_1^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{\text{p}}] + \\ & \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_d^{\text{p}}] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{\text{i1}}] \delta_{i1i2} - \\ & \frac{3}{16} m_2 \tilde{\mu} g_2^2 (\overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} + \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}}) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{\text{i2}}] + \\ & \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_u^{\text{p}}] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,1,1,0} [m_3, m_d^{\text{p}}, m_q^{\text{i2}}] - \\ & \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_3, m_d^{\text{p}}, m_q^{\text{i2}}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_3, m_d^{\text{p}}, m_q^{\text{i2}}] + \\ & \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_3, m_q^{\text{i2}}, m_d^{\text{p}}] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,1,1,0} [m_3, m_q^{\text{i2}}, m_u^{\text{p}}] + \\ & \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_3, m_q^{\text{i2}}, m_u^{\text{p}}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_3, m_q^{\text{i2}}, m_u^{\text{p}}] - \\ & \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_3, m_u^{\text{p}}, m_q^{\text{i2}}] + \frac{1}{16} \tilde{\mu} \overline{y_d^{\text{i2p}}} \overline{y_u^{\text{rs}}} y_u^{\text{i1s}} a_d^{\text{rp}} \\ & \text{LF}_{2,2,1,-1} [m_d^{\text{p}}, \tilde{\mu}, m_q^{\text{r}}] - \frac{1}{16} \tilde{\mu} \overline{y_d^{\text{pr}}} y_d^{\text{i1r}} \overline{y_u^{\text{i2s}}} a_u^{\text{ps}} \text{LF}_{2,2,1,-1} [m_q^{\text{p}}, \tilde{\mu}, m_u^{\text{s}}] - \\ & \frac{1}{16} \tilde{\mu} \overline{y_d^{\text{i2p}}} \overline{y_u^{\text{rs}}} y_u^{\text{i1s}} a_d^{\text{rp}} \text{LF}_{2,2,1,-1} [m_q^{\text{r}}, \tilde{\mu}, m_d^{\text{p}}] + \frac{1}{16} \tilde{\mu} \overline{y_d^{\text{pr}}} y_d^{\text{i1r}} \overline{y_u^{\text{i2s}}} a_u^{\text{ps}} \\ & \text{LF}_{2,2,1,-1} [m_u^{\text{s}}, \tilde{\mu}, m_q^{\text{p}}] + \frac{1}{24} m_1 \tilde{\mu} g_1^2 (\overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} - \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}}) \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_q^{\text{i2}}] + \\ & \frac{1}{24} m_1 \tilde{\mu} g_1^2 (-\overline{y_d^{\text{i2p}}} y_d^{\text{i1p}} + \overline{y_u^{\text{i2p}}} y_u^{\text{i1p}}) \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_q^{\text{i2}}] + \\ & \frac{3}{8} m_2 \tilde{\mu} g_2^2 (\overline{y_d^$$